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Universidad Politécnica de Madrid
ETS de Ingeniería de Sistemas Informáticos

# Metodologías de Desarrollo Web

Máster de Ingeniería Web
E.T.S. de Ingeniería en Sistemas Informáticos
Universidad Politécnica de Madrid
Curso 2025-2026



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Universidad Politécnica de Madrid
E.T.S. de Ingeniería en Sistemas Informáticos
Departamento de Sistemas Informáticos

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- **CONTENIDOS**
  - **Conceptualización de product**
  - **Inception**
  - Roadmapping
  - Gestión del backlog
  - Escritura de historias de usuario



# Introducción a la Conceptualización





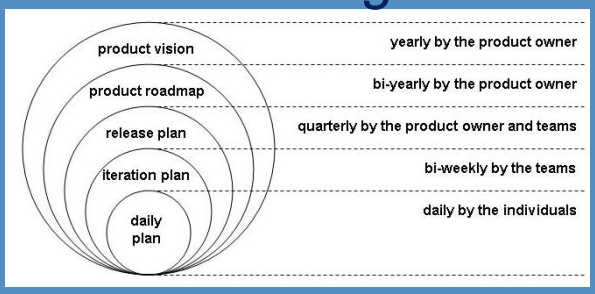


# Niveles de "*agilidad*"

Abstracto



Concreto




**Visión del producto (nivel 1)**
- Indica como deberá ser una organización o un producto en el futuro
- Indica los elementos necesarios del producto y los esfuerzos para conseguirlo

**La hoja de ruta (nivel 2)**
- Indica la forma en la que va a ser producido el producto
- Indica las etapas en el "viaje" hacia la construcción del producto

**Release Planning (nivel 3)**
- Agrupación de características del producto
- Determina incrementos que aportan valor al cliente

**Iteration Planning (nivel 4)**
- Aumenta el detalle de las características del producto
- Disminuye la incertidumbre

**Daily Plan (nivel 5)**
- Sesión de seguimiento del proyecto



## Product or Project

What business objectives will the product fulfill?

Product Goals

Product Charter

Customers

User Personas

## Iteration or Sprint

What specifically will we build? (user stories)

How will this iteration move us toward release objectives?

Iteration Goal

Development or Construction Tasks

## Release

How can we release value incrementally?

What subset of business objectives will each release achieve?

What user constituencies will the release serve?

What general capabilities (big stories) will the release offer?

Release Roadmap

Target Customers

Target Personas

## Story (Backlog Item)

What user or stakeholder need will the story serve?

How will it specifically look and behave?

How will I determine if it's completed?

Story Details

Acceptance Tests

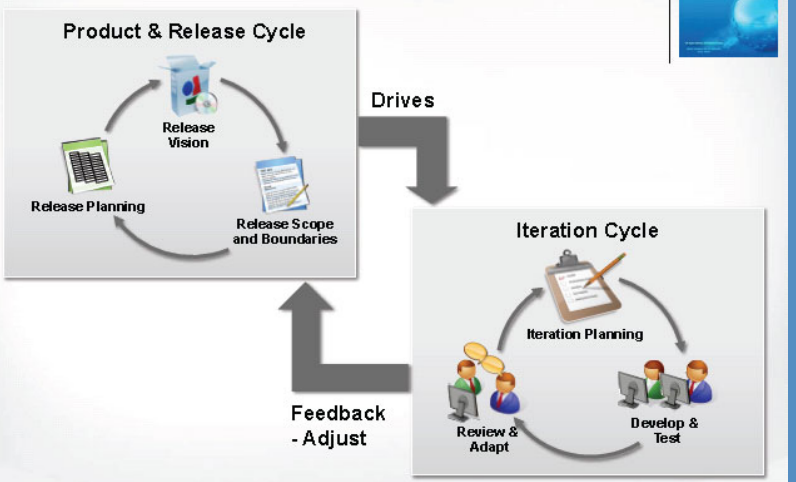
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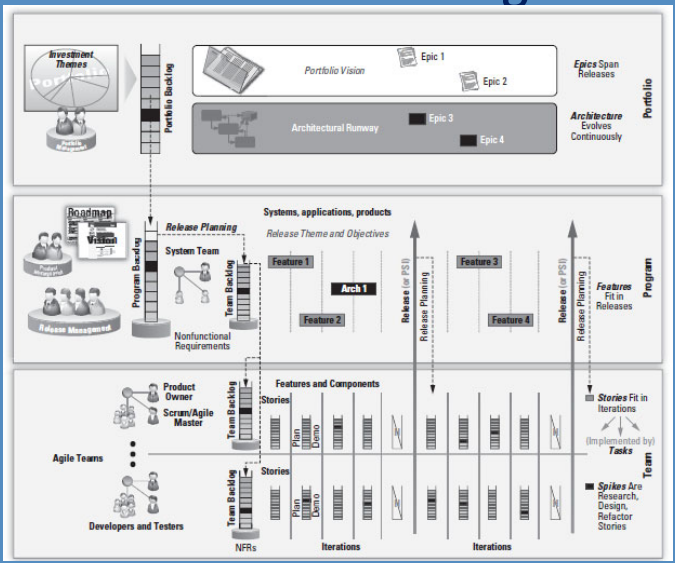
## Two Level Planning and Tracking




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# Niveles de "*agilidad*"



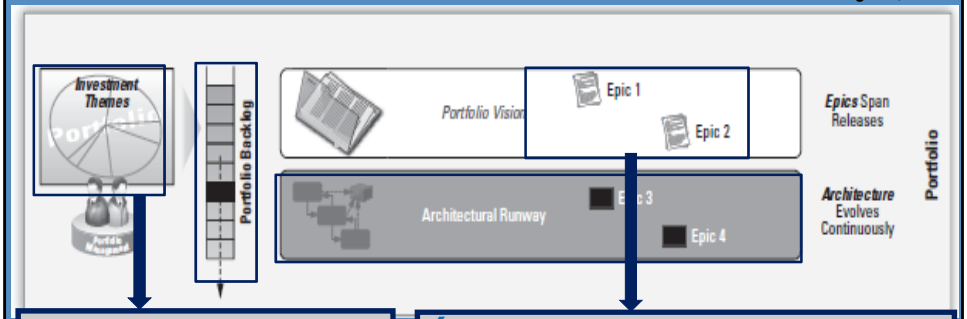

***Agile Software Requirements,***
*Dean Leffingwell, 2011*





# Niveles de "*agilidad*"
## Portafolio

***Agile Software Requirements,***
*Dean Leffingwell, 2011*



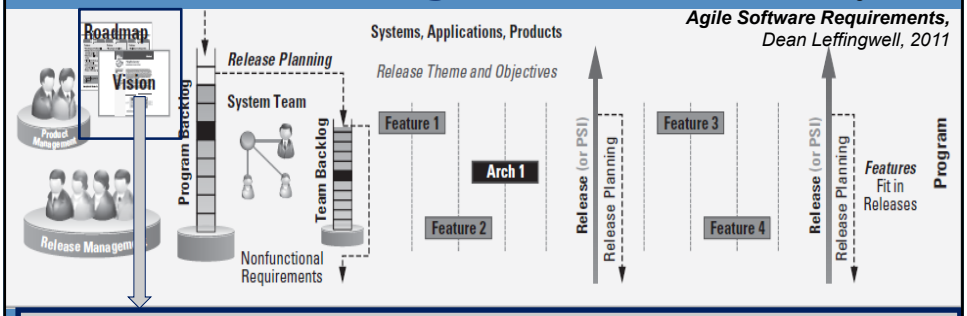

**Temas de Inversión** Objetivo de inversión de la empresa o unidad de negocio

**Épicas** Necesidades del cliente expresadas al mayor nivel de abstracción.
Iniciativas de desarrollo para dar valor a los temas de inversión

# Niveles de "*agilidad*"

## Programa

***Agile Software Requirements,*** *Dean Leffingwell, 2011*




***Agile Software Requirements,*** *Dean Leffingwell, 2011*

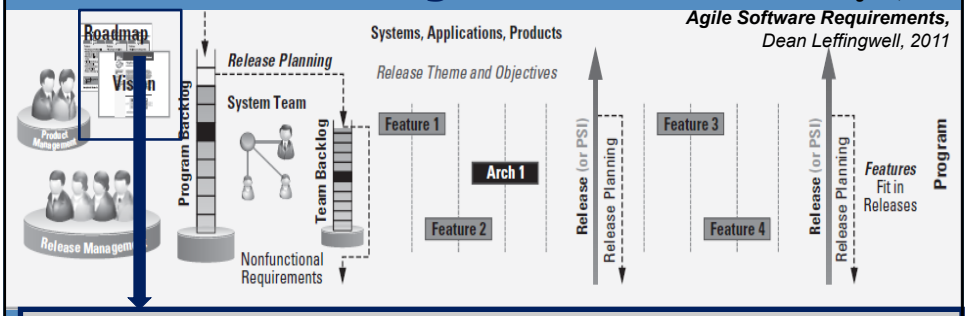
**Visión** Del producto en el dominio de aplicación en Features. Contesta a las preguntas esenciales:¿Qué problema resuelve? ¿Qué beneficios y propiedades proporciona? ¿Para quién es? ¿Que rendimiento, fiabilidad, etc., proporciona? ¿Qué plataformas, estándares, aplicaciones, etc. soportará?

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# Niveles de "*agilidad*"

## Programa

***Agile Software Requirements,*** *Dean Leffingwell, 2011*

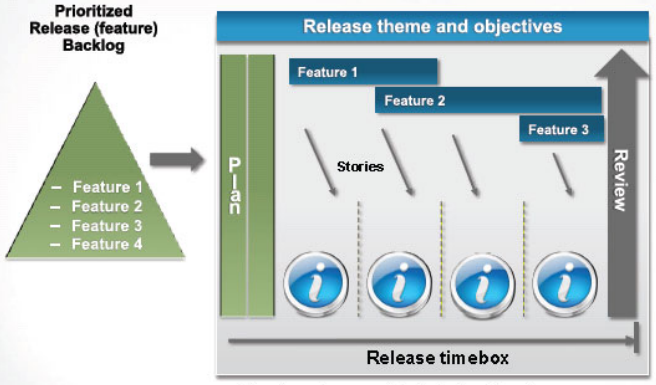



***Agile Software Requirements,*** *Dean Leffingwell, 2011*

**Roadmap** Planificación temporal de las releases. Establece para cada reléase:
-Fecha de entrega
-Objetivos y tema de la release
-Conjunto de features priorizada

## Release Pattern



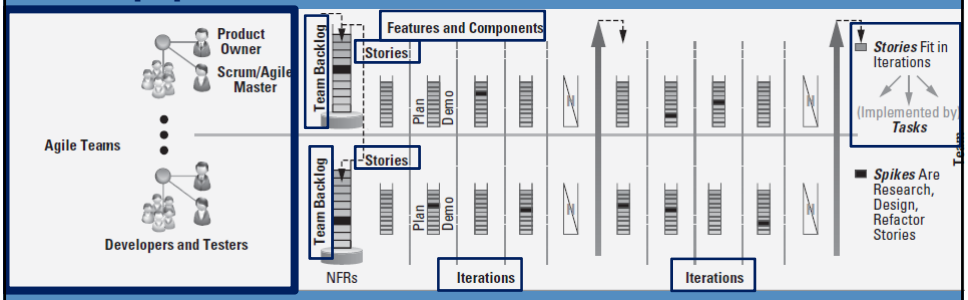

- Fixed cadence with fast feedback
- Actively manage release queues.
- Small, global batch sizes. Limit global work in process.
- Forced global synchronization
- Low global transaction and experimentation costs

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## Niveles de "*agilidad*"

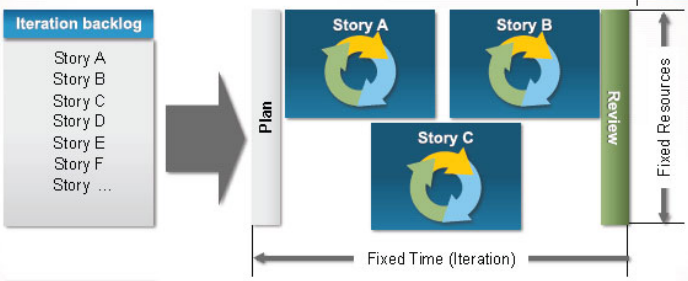
### Equipo




***Agile Software Requirements,***
*Dean Leffingwell, 2011*

## Iteration Pattern




Define
Develop
Accept

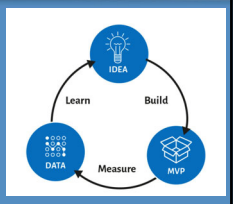
- Fixed cadence with fast feedback
- Actively manage queues. Small, local batch sizes. Limit work in process.
- Forced local synchronization
- Low transaction and experimentation costs

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# About Inception

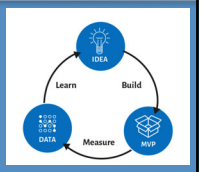



- What is Inception?
  - It is the set of activities conducted by a project team before to start the delivery work.
  - It could take days or weeks depending on the project size and the available knowledge.
  - It removes everything that is not about the product
  - The result is the MVP of the product.

Inception

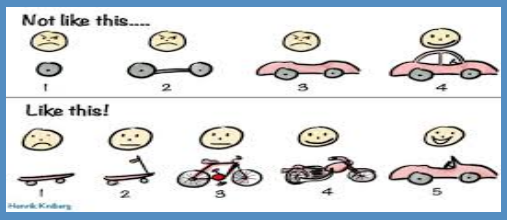
- **Vision**
- **Is - Is not**
- **Personas**
- **Features**
- **Architecture**
- **User journey**
- **Roadmapping**
- **Team building**

# About MVP




- MVP = Minimum Viable Product
  - A simple version of a producto that is given to users in order to validate the key business assumptions.
  - Product is build incrementally.
  - The idea behind MVP is product enhancement guided by proven hypothesis validation.
  - Each MVP increment is an evolutioned version of the solution.

Think big!
Start small &
Learn fast!



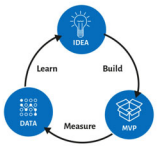

~~MVP~~

**MVP**

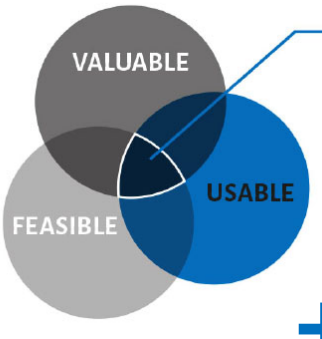




# About MVP




Valuable: in terms of commercial value and return of investment.




Usable: tailored according the needs, desires and limitation

Feasible: there are technology and knowledge to elaborate it

\+

THE "WOW" FACTOR

# 2. Preparando el trabajo





# Bases para la creación del Product Backlog

- Workshop de creación del Product Backlog
  - Proyectos pequeños → 1 ó 2 días
  - Proyectos grandes → Los días necesarios
  - Conocido como “release planning” en Scrum Guide
  - Se realiza una única vez al inicio del proyecto
  - El product backlog se mantiene de forma continua durante el proyecto
  - Se realizan múltiples actividades
    - Definir el producto
    - Identificar sus características

# Workshop para creación del backlog

- Participantes
  - Product Owner Team
  - Development Team memb
  - Workshop facilitator



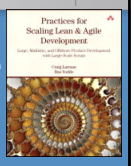




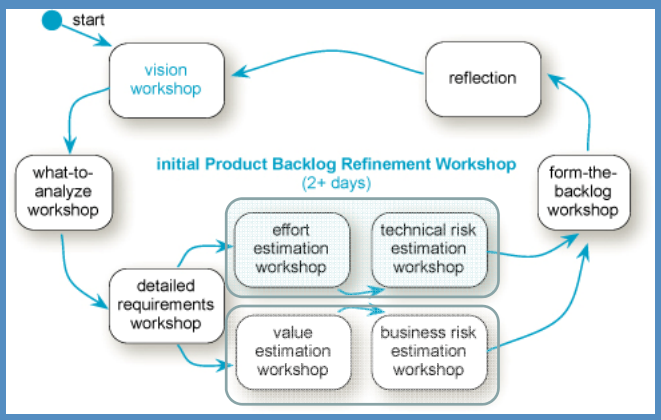




# Workshop para creación de backlog




- Activities




DevTeam

PO Team

# Workshop para creación del backlog

- Formato
  - Es un taller de trabajo
  - Participativo
  - Tipos:
    - World café
    - Fish bowl
    - Brainstorming







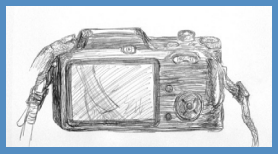




# Workshop para creación del backlog

- Tools





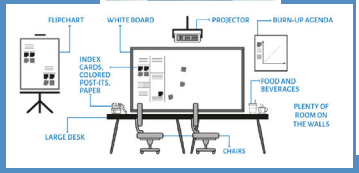


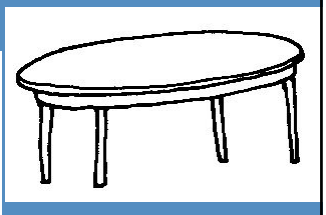










# Workshop para creación del backlog



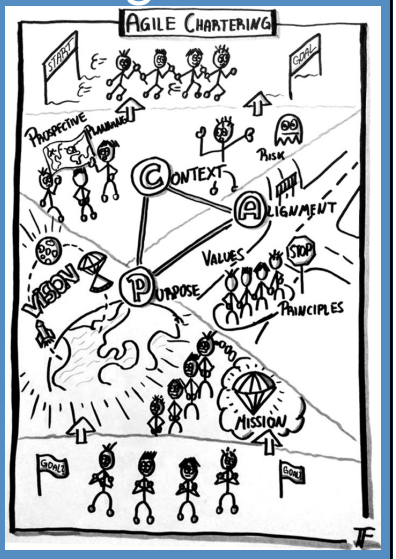







# Project Chartering

- A project charter is a high-level summary of the project's key success factors, synthetic enough that it can be displayed on one wall of the team room as a flipchart-sized sheet of paper.
- It includes at least
  - **Purpose**: vision, mission, mission tests
  - **Context**: scope boundaries, how and whom and risks
  - **Alignment**: and reciprocal agreements between the project's implementation team and external stakeholders.

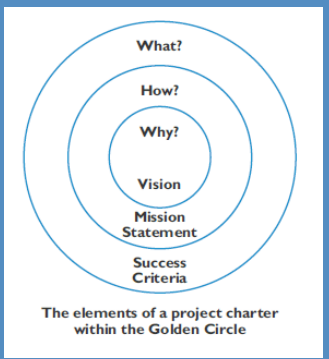







# Project chartering

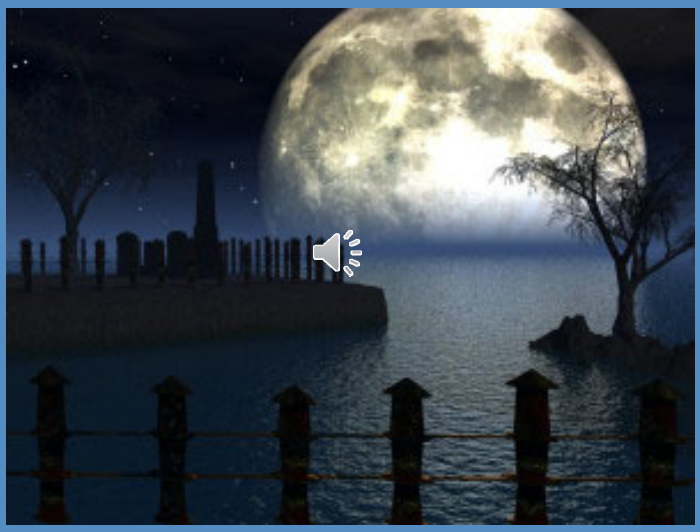
- Main goals
  - ***Why*** are we building this product?
  - ***What*** are we building?
  - ***How*** will we know if it is successful?
  - ***Who*** is the project community?
- Outcomes
  - A common understanding of the product,
  - Its vision,
  - Its goals.
  - Success criteria



The elements of a project charter within the Golden Circle

This leads to a higher level of engagement for the team members and provides a richer context for all involved.

# Vision and Mision – Tale of the







# Project Chartering

- **Vision**
  - An Agile project charter vision is a simple call to action.
  - It answers questions for the system/product to be built.
  - Enables to know the essence of the system/product.
  - It boots and drives the team.

"The vision describes why the project is being undertaken and what the desired end state is" (Ken Schwaber, 2004, 68)

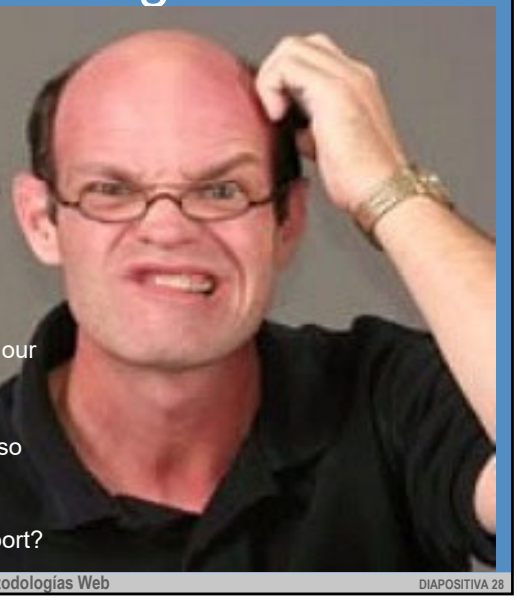


sharoncm.blogspot.com

# Project Chartering - Vision

Vision should answer the following questions:

- What problem does this particular solution solve?
- What features and benefits does it provide?
- For whom does it provide it?
- What are the main challenges and risks?
- What are the main benefits?
- What are the main diferences with our competitors?
- What is our time to market slot?
- What performance, reliability, and so on, does it deliver?
- What platforms, standards, applications, and so on, will it support?



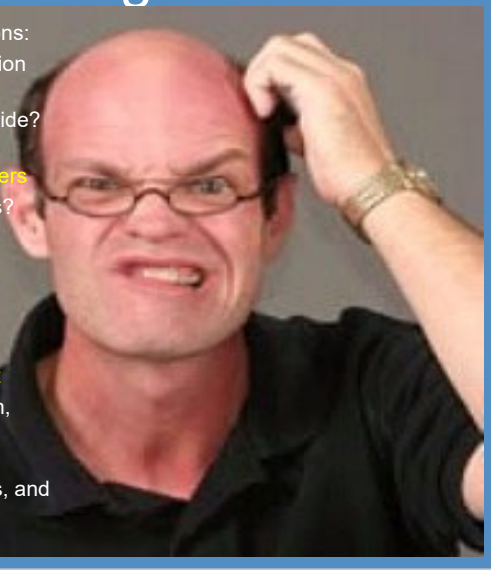




# Project Chartering - Vision

Vision should answer the following questions:

- What problem does this particular solution solve? Goals
- What features and benefits does it provide? Functionalities
- For whom does it provide it? Stakeholders
- What are the main challenges and risks? Risks
- What are the main benefits? Goals and Market
- What are the main diferences with our competitors? Market
- What is our time to market slot? Market
- What performance, reliability, and so on, does it deliver? Non Functional Requirements (-ilities and constraints)
- What platforms, standards, applications, and so on, will it support? Non Functional Requirements (-ilities and constraints)



# Project Chartering



- **Mission**
  - This is the "What" of the project
  - It states what will be done in the project to achieve its higher purpose. "How will we work together to achieve this vision?"
    - Example:
      - To work collaboratively to make decisions with the project owner
      - To communicate daily about progress and barriers.
      - To collaborate and communicate in real-time whenever possible.
      - To perform work in established timeboxes (planning meeting, work period, review meeting and lessons-learned meeting.
      - To determine next steps prior to the planning meeting.





# Project Chartering

- **Success criteria**
  - How will we know when we've achieved it … or when it needs to change?
  - The success criteria outline the outcomes the project will achieve.
  - They must be feasible and demonstrable



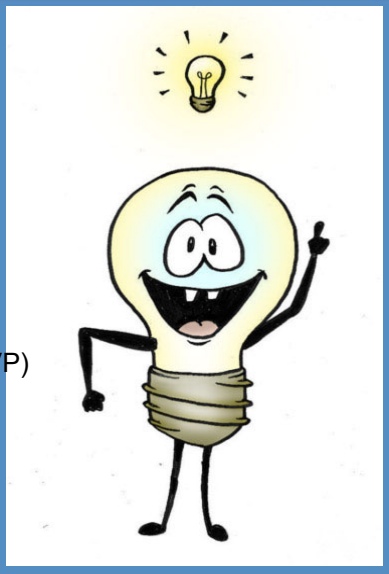
https://lamenteesmaravillosa.com/wp-content/uploads/2014/05/mujer-exitosa.jpg

# Project Chartering - Vision

Product Vision should:

- be shared and unified
- be general and attractive
- be small and nice
- be simple
- represent Minimum Viable Product (MVP)

How to get the Vision?



http://1.bp.blogspot.com/LightbulbIdea.jpg





# Inception deck



| | | |
|---|---|---|
| Why are we here? | Elevator pitch | Product box |
| NOT list | Meet the neighbors | Show solution |
| Up at night | Size it up | What's going to give |
| | What's it going to take | |

# Inception Deck





- Why are we here?
  - Understand what we are building
  - Why
    - Make better, more informed decisions
    - Balancing the conflicting forces and trade-offs
    - Come up with innovative solutions because they are
  - Feature-based



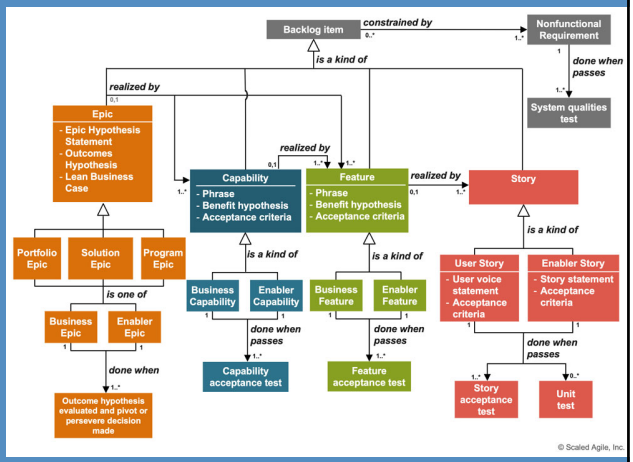


# Inception Deck

- Why are we here?

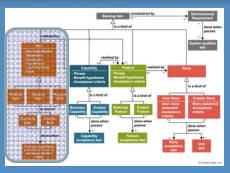
Requirements model

Epic
Capability/Enabler
Feature
Story




https://stagingv5.wpenginepowered.com/wp-content/uploads/2020/12/SAFe_Requirements_Model_F01_WEB.png

# Inception Deck






- Why are we here?
- Epic
  - It is a container for a significant Solution development initiative that captures the more substantial investments.
  - Epics require the definition of a Minimum Viable Product (MVP)
  - Two types:
    - Business epics directly deliver business value,
    - Enabler epics are used to advance the Architectural Runway to support upcoming business or technical needs.

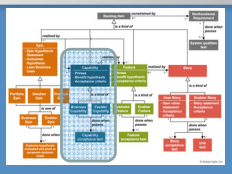
| Epic Hypothesis Statement | |
|---|---|
| **Funnel Entry Date:** | <The date that the epic entered the funnel.> |
| **Epic Name:** | <A short name for the epic.> |
| **Epic Owner:** | <The name of the epic owner.> |
| **Epic Description:** | <An elevator pitch (value statement) that describes the epic in a clear and concise way.><br><br>**For** <customers><br>**who** <do something><br>**the** <solution><br>**is a** <something - the 'how'><br>**that** <provides this value><br>**unlike** <competitor, current solution or non-existing solution><br>**our solution** <does something better – the 'why'> |
| **Business Outcomes:** | <The measurable benefits that the business can anticipate if the epic hypothesis is proven to be correct.> |
| **Leading Indicators:** | <The early measures that will help predict the business outcome hypothesis. For more on this topic, see the Innovation Accounting advanced topic article.> |
| **Nonfunctional Requirements (NFRs):** | <Nonfunctional requirements (NFRs) associated with the epic.> |

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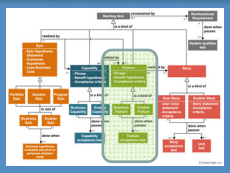
# Inception Deck






- Why are we here?
- Capability
  - It is a higher-level solution behavior.
  - It may describe as abilities or qualities that can be created or used to achieve desired outcomes.
  - They are too big to be delivered.
  - Capabilities are sized and split into multiple features to facilitate their implementation.
- Enabler
  - It is a technical ability of the platform that supports the expansión of business and crucial for driving business features.
  - They are used for exploration, evolving the architecture, improving architecture and compliance activities.

# Inception Deck



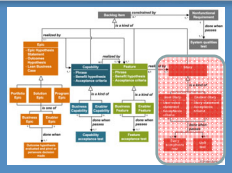



- Why are we here?
- Feature
  - It is a service that fulfills a stakeholder need.
  - Features may also be described as parts of a product that provide notable value to the user.
  - They add some substance to the vision when sharing this with teams and stakeholders.
  - Each feature includes a benefit hypothesis and acceptance criteria, and is sized or split as necessary to be delivered.
  - Features typically provide functionality for multiple user roles.
  - Two types: business (PO) and enablers (Architects)





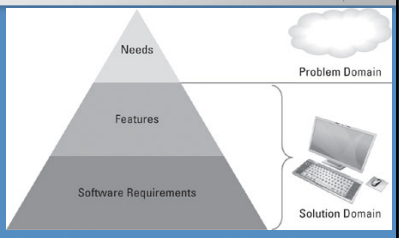
# Inception Deck






- Why are we here?
- Story
  - Stories are short descriptions of a small piece of desired functionality, written in the user's language.
  - It represents small, vertical slices of system functionality and are sized to be completed in a single Iteration.
  - Stories are the primary artifact used to define system behaviour.
  - It provides just enough information for both business and technical people to understand the intent.
  - Stories are first written on an index card or sticky note following a predefined template.

# Inception Deck
# Why are we here?




* Or any other name
  - Levels of abstraction

Needs
- Epic Themes*: establish the relative investment objectives for the enterprise or business unit.
- Capabilities *: represent the highest-level expression of a customer need.
- Features *: describe what new things the system will do for its users and the benefits the user will derive
- Stories *: describe functional system behaviour:
  - Epic Stories *: functionality that is still too complex to be implemented.
  - User Stories *: are what agile teams use to define the system behavior and value for the user.



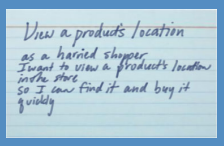


# Inception Deck – Why are we here?

| Epic | Capabilities | Features | Epic Story | User Story |
|---|---|---|---|---|
| Shared mobility | Car hire | Hiring management | Hire a car | Online |
| | | | | Mobile |
| | | | Drop-off a car | Online |
| | | | Queries | By date |
| | | | | By car |
| | | Car localization | … | … |
| | | Data Analytics | … | … |
| | Bike hire | … | … | … |
| | Scooter hire | … | … | … |

# Inception Deck

## How large are features?




- Think about the same level of abstraction
- If it is not possible …
- Think about the same level of abstraction

- Think in terms of product features and capabilities
- Thing big, starts small and learn fast!









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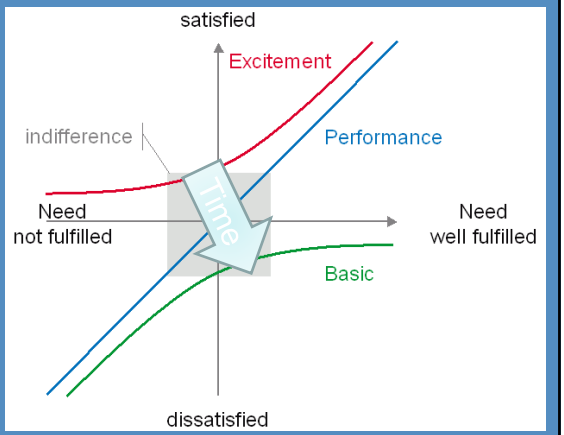




# Inception Deck

## El modelo de Kano

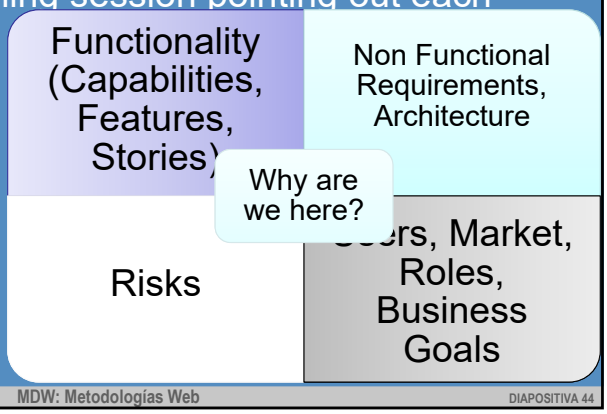
- Atributos básicos
  - Son obligatorios. Si no se tienen se rechaza. Umbral mínimo
- Lineales
  - Causan felicidad si se tienen y nada si no se tienen
- Entusiasmo
  - Causan felicidad si se tienen pero no se tiene constancia de ellas hasta que se presentan
- Indiferentes
- Cuestionables
- Inversas



# Inception Deck

- Why are we here?
  - Run a world café with (at least) four corners
  - Run brainstorming session pointing out each topic

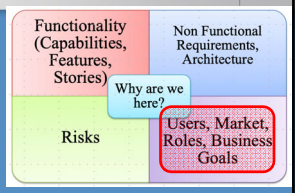
1 idea
=
1 Sticky note








# Inception Deck – Why are we here?

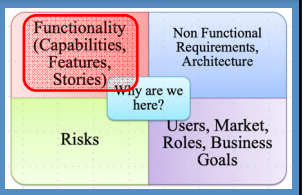



- Users, Market, Roles and Business corner
  - Identify:
    - Types of users
    - Types of roles for each user
    - Market opportunities
    - Stakeholders
    - Business Goals
  - Prioritize Business Goals
  - Classify market opportunities by relevance
  - Associate stakeholders
  - Associate users with market opportunities

**Personas HCI practice**



# Inception Deck – Why are we here?



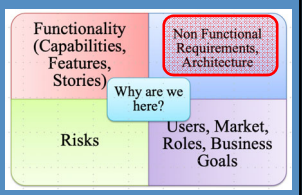

- Functionality
  - Identify:
    - Capabilities
    - Features
    - Stories

    **Any user need**
  - Classify needs (group by similarities)
  - Associate need with market needs
  - Associate needs with Personas
  - Associate needs with stakeholders



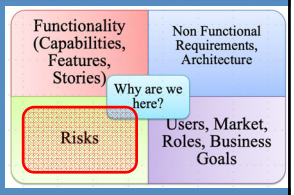


# Inception Deck – Why are we here?




- Non-functional requirements
  - Identify:
    - Any constraint impacting the product
    - Any technology that should be considered
    - Any architectural requirements
    - Any platform constraint
  - Prioritize non-functional requirements

# Inception Deck – Why are we here?




- Risks
  - Identify risks:
    - About technology
    - About development
    - About teams
    - About market
    - About competitors
  - Prioritize Risks
  - Associate risks with non-functional requirements (if possible)
  - Associate risks with stakeholders (if possible)



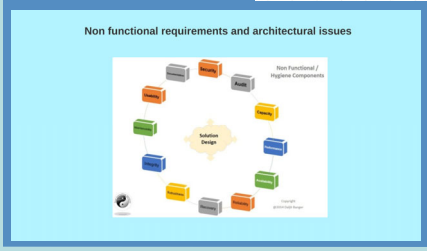


# Inception deck




- Why are we here?









# Inception deck

- Why are we here?

| | |
|---|---|
| Write items in sticky notes | One sticky note only one idea |
| Assign sticky notes to roles | Categorize sticky notes |





# Inception Deck- Elevator Pitch




- Crear elevator pitch.
  - Breve descripción del problema (45 seg)
  - Resaltar SOLO la idea para fijar el proyecto
  - Pensar en el cliente
  - Tiene que contener los siguientes temas:
    - Para quién
    - Que tiene una necesidad
    - El nombre del producto
    - Describir de qué tipo es
    - Describir lo que hace
    - Indicar lo que hace la competencia
    - Ventajas de nuestro producto



The fumbled response

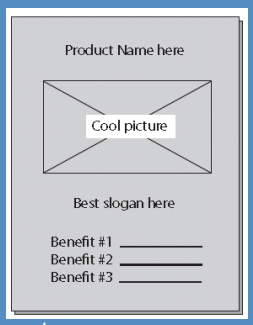
**The Elevator Pitch Template**

- For [target customer]
- who [statement of need or opportunity]
- the [product name]
- is a [product category]
- that [key benefit, compelling reason to buy].
- Unlike [primary competitive alternative]
- our product [statement of primary differentiation].

# Inception Deck- Product Box

- Diseño del Product Box
  - Tormenta de ideas de los beneficios
    - Características -> Beneficios
  - Buscar un eslogan
    - Corto
    - Directo
    - Emotivo
  - Crear el Product Box
    - Dibujo
    - Eslogan
    - Las características más relevantes

Front




Back

- Feature 1
- Feature 2
- Feature 3

…

- Feature N





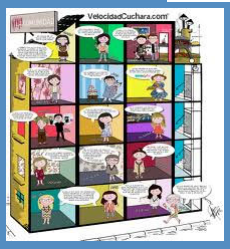
# Inception Deck- Not List

- Create a NOT list
  - Manejar tres secciones
    - En el ámbito del proyecto
    - Fuera del ámbito del proyecto
    - No está claro
  - Objetivo tener claro el foco del proyecto

| In Scope | Out of Scope |
|---|---|
|  |   |
| Indeterminado | |
|    | |

# Inception Deck

- Meet your neighbors
  - No estamos solos en la empresa
  - Identificar las personas relacionadas con el proyecto
  - Están fuera pero tenemos dependencias con ellos
  - Tipo de relación





Café
(Te diviertes con ellos)



Donuts
(Aprecias su colaboración)




Confianza
(Los valoras)

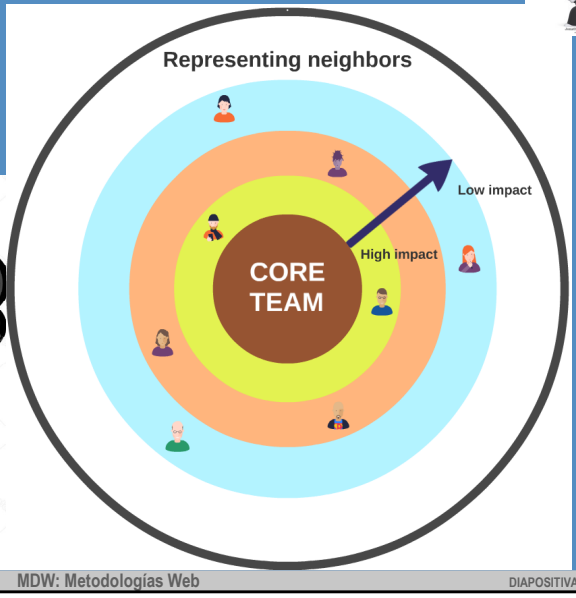




# Inception Deck

## Meet your neighbors

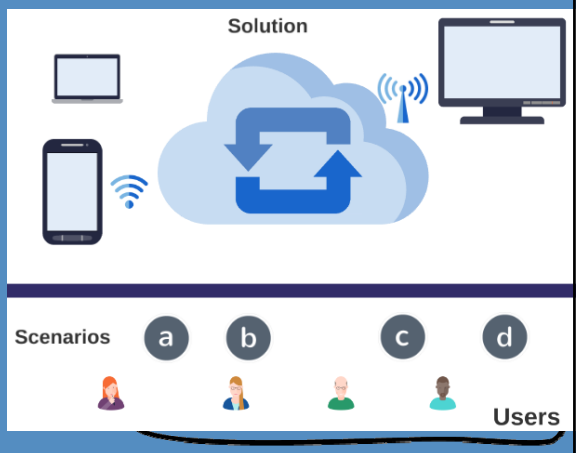


# Inception Deck



- Show your solution
  - Visualizar la solución
  - Mostrar elementos de la solución
    - Arquitectura
    - What-if escenarios
    - Tamaño del sistema
  - Mostrar los límites y ámbito
  - Mostrar los riesgos
  - Representa el acuerdo
  - NO es DETALLADO



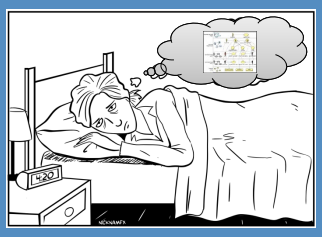


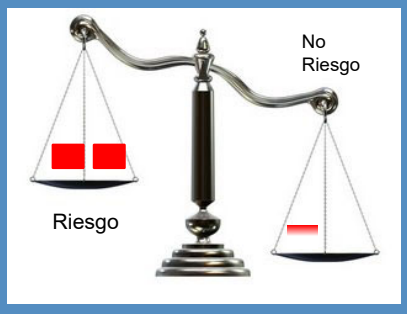



# Inception Deck



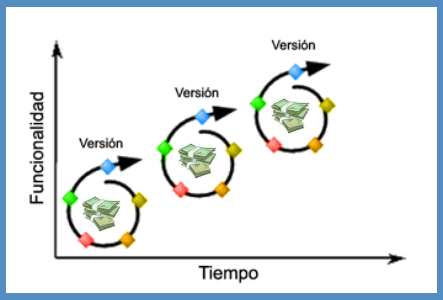
- Keep us Up at night
  - Siempre somos optimistas … pero no dormimos
  - Siempre hay más cosas que hacer que tiempo y dinero
  - Retos
  - Riesgos y no riesgos





# Inception Deck

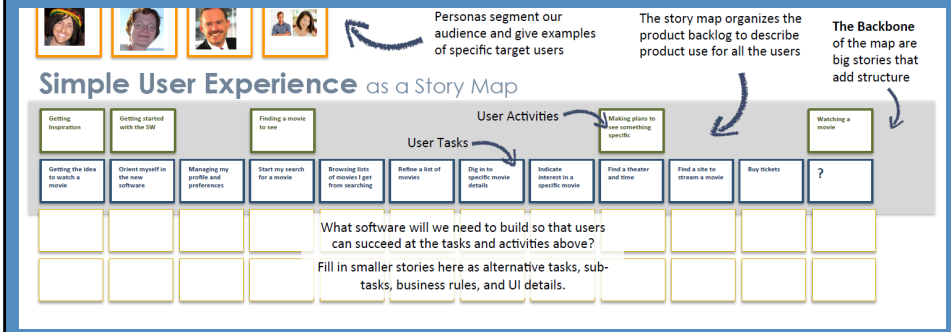
- Size it up
  - Conocer el horizonte temporal
    - Corto, medio y largo plazo
  - Expectativas de los stakeholders
  - Grano grueso
  - Roadmapping








# Inception Deck – Size it up



# Inception Deck
## ¿Qué es un hoja de ruta - roadmap?

**What is a Roadmap?**

*"It is a plan that matches short-term and long-term goals with specific solutions to help meet those goals"*

It is a living document
Answer key strategic questions







# Inception Deck
## El roadmap …

- … permite al equipo capturar los objetivos de las siguientes versiones
- … da un sentido a como una empresa espera los incrementos de valor a lo largo del tiempo.
- … es un artefacto de planificación a través de sus versiones.
- … siempre centrada en la esencia del producto




https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcS5TsQS3gS_b5ZvJEK33D0MwDhBWA5cP2IE1SEQfYtk_BO4mzwE

# Inception Deck





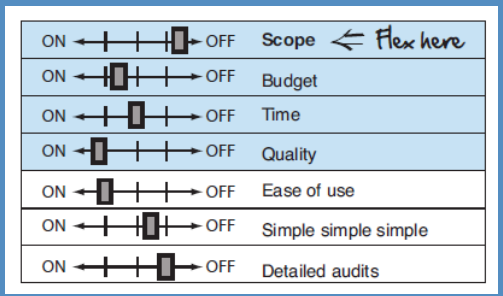



# Inception Deck




- What's Going to Give?
  - Factores que influyen
  - Buscar equilibrio
  - Resolver conflictos
  - NO todos ON
  - NO todos OFF
  - NO dos al mismo nivel
  - Cantidad limitada



# Inception Deck

- What it is going to take?
  - Determinar el equipo A
  - Identificar claramente al PO
  - Estimar el coste














# Agile Chartering - Summarizing



Purpose

Alignment

Context

Why are we here?

Elevator pitch

Product box

NOT list

Meet the neighbors

Show solution

Up at night

What's going to give

What's it going to take

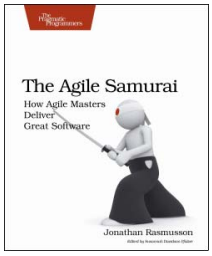
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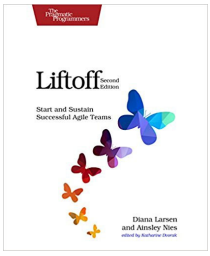






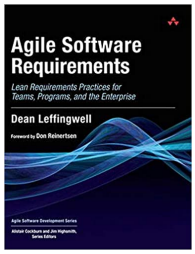
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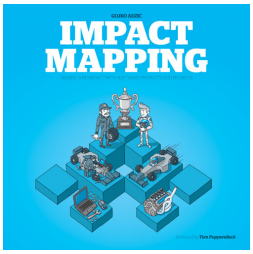












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